

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

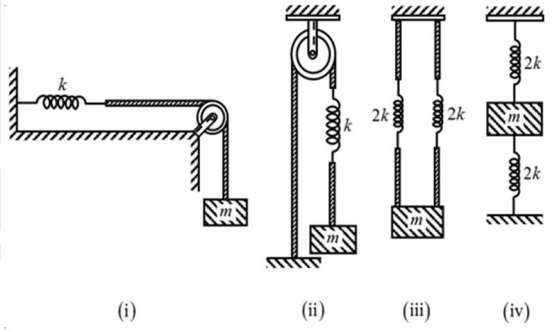
Sec: SR STAR - A & APEX, ALL IITS **DT: 30 /03 /2026**
Time: 3 Hrs **GTM - 09** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A block of mass *m* is suspended by different springs of force constant as shown in the figure.



Let the time period of oscillation in these four positions be $T_1, T_2, T_3 \text{ and } T_4$. Then

A) $T_1 = T_2 = T_4$

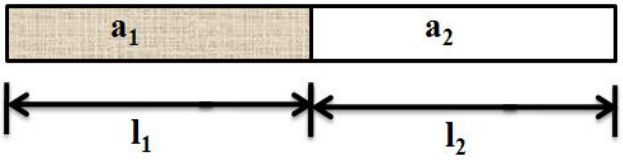
B) $T_1 = T_2 \text{ and } T_3 = T_4$

C) $T_1 = T_2 = T_3$

D) $T_1 = T_3 \text{ and } T_2 = T_4$

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2. Two rods having lengths l_1 and l_2, made of materials with linear expansion coefficients α_1 and α_2, were soldered together. The equivalent coefficients of linear expansion for the composite rod is



A) $\frac{l_1\alpha_2+l_2\alpha_1}{l_1+l_2}$ B) $\frac{l_1\alpha_1+l_2\alpha_2}{\alpha_1+\alpha_2}$

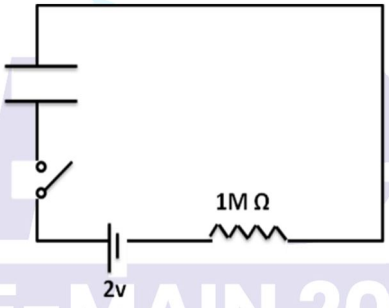
C) $\frac{l_1\alpha_1+l_2\alpha_2}{l_1+l_2}$ D) $\frac{l_2\alpha_1+l_1\alpha_2}{\alpha_1+\beta_2}$

3. A metal ball immersed in water weighs W_1 at 0^oC and W_2 at 50^oC. The coefficient of cubical expansion of metal is less than that of water. Then

A) $W_1 > W_2$ B) $W_1 < W_2$

C) $W_1 = W_2$ D) Data is insuifficient

4. A parallel plate capacitor with circular plates of radius 0.5m has a capacitance of $1nF$. At t=0 it is connected for charging in series with a resistor R = 1 MΩ across a 2V battery as shown in figure. The magnetic field at a point P, which is at a distance of 0.25 m form the axis of the plates, at $t = 10^{-3}$ s is (approximately).



A) 1.48×10^{-13} T B) 0.74×10^{-13} T C) 0.74×10^{-7} T D) 1.48×10^{-7} T

5. When a metallic surface is illuminated with light of wavelength λ, the stopping potential is V. When the same surface is illuminated by light of wavelength 2 λ, the stopping potential is V/3. The threshold wavelength for the surface is

A) $\frac{4\lambda}{3}$ B) 4λ C) 6λ D) $\frac{8\lambda}{3}$

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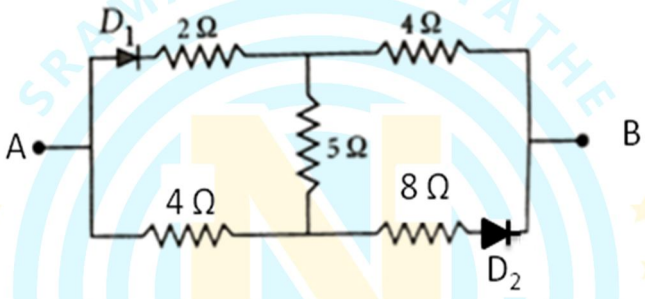
6. Consider the nuclear reaction

$X^{200} \rightarrow A^{110} + B^{90}$

If the binding energy per nucleon for X, A and B is 7.4 MeV, 8.2 MeV and 8.2 MeV respectively, what is the energy released?

A) 200 MeV B) 160 MeV C) 110 MeV D) 90 MeV

7. The equivalent resistance of the circuit across AB is given by



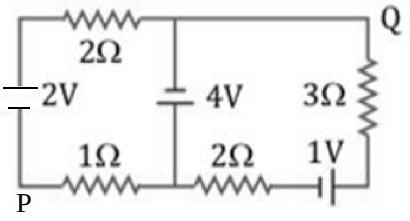
A) 4 Ω B) 13 Ω

C) 4 Ω or 13 Ω D) 4 Ω or zero

8. A particle of mass 'm' is moving with speed $2v$ and collides with a mass '2m' moving with speed v in the same direction. After collision, the first mass is stopped completely while the second one splits into two particles each of mass m, which move at an angle 45° with respect to the original direction. The speed of each of the moving particle will be

A) $\sqrt{2}v$ B) $\frac{v}{\sqrt{2}}$ C) $\frac{v}{(2\sqrt{2})}$ D) $2\sqrt{2}v$

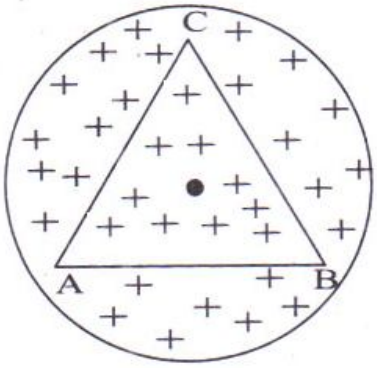
9. In the circuit shown, what is the potential difference V_{PQ}?



A) $+3V$ B) $+2V$ C) $-2V$ D) None

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10. A triangular wire frame having each side equal to 2m is placed in a region of time varying magnetic field having $dB/dt = \sqrt{3}T/s$. The magnetic field is perpendicular to the plane of the triangle as shown in figure. The base of the triangle AB has a resistance 1Ω while the other two sides have resistance 2Ω each. The magnitude of potential difference between the points A and B will be



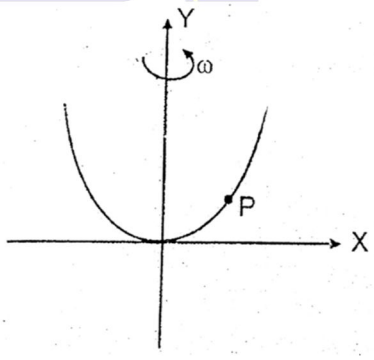
A) 0.4 V B) 0.6 V C) 1.2 V D) None

11. Equal amount of water and ice are mixed. The ice is at temperature $-10^{\circ}C$ and water is at $40^{\circ}C$. Assuming no heat exchange between system and surrounding, the fraction of ice melted is___

(given $L_f = 80\,cal/g, S_{ice} = 0.5\,cal/g^0C, S_{water} = 1\,cal/g^0C$)

A) 7/16 B) 5/6 C) 5/16 D) 9/16

12. A curved smooth wire as shown in diagram is revolving with constant angular velocity about y-axis, a bead in this wire remains in equilibrium at every position then shape of wire:



A) May be straight line B) Parabolic

C) $y \alpha x^4$ D) Not possible for any shape

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13. Two vertical parallel rectangular glass plates are partially submerged in water. The separation between the plates is d=0.1 mm and the width of each plate is $\ell = 10\,cm$. Assuming that the water between the plates does not reach the upper edges of the plates and the contact angle is $\theta = 0^o$. The surface tension of water is $T = 7\times10^{-2}\,N/m$ and the density of water is $\rho = 10^3\,kg/m^3$. Then choose the correct option. $\left(g = 10\,m/s^2, atmospheric\ pressure = 10^5\,Pa\right)$

A) The water rises up to height h=16 cm between the plates.

B) The water rises up to height h=28 cm between the plates

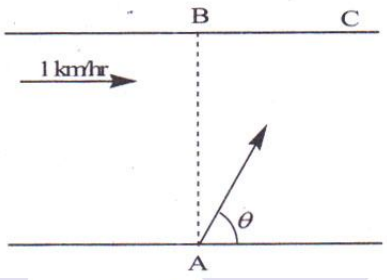
C) The force of mutual attraction between the plates is 4.9 N

D) The force of mutual attraction between the plates is 9.8 N

14. If time (t), velocity (v), and angular momentum (ℓ) are taken as the fundamental units. Then the dimension of mass (m) in terms of t, v and ℓ is

A) $\left[t^{-1}v^{1}\ell^{-2}\right]$ B) $\left[t^{1}v^{2}\ell^{-1}\right]$ C) $\left[t^{-2}v^{-1}\ell^{1}\right]$ D) $\left[t^{-1}v^{-2}\ell^{1}\right]$

15. A river is flowing with a speed of 1 km/hr. A swimmer wants to go to point 'C' starting from 'A'. He swims with a speed of 5 km/hr, at an angle θ w.r.t the river. If AB=BC=400m. Then the value of θ is



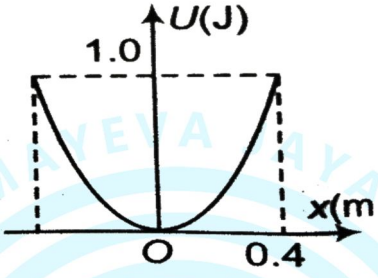
A) 37° B) 30° C) 53° D) 45°

16. A light beam travelling in +ve x-direction is described by $E_y = 300\sin\left(\omega t - \frac{\omega x}{c}\right)V/m$. An electron is constrained to move along y direction with a speed of $2\times10^7\,ms^{-1}$. The maximum magnetic force (in Newton) on the electron is

A) 3.2×10^{-18} B) 5.1×10^{-16} C) 6.5×10^{-11} D) 7.8×10^{-12}

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17. A particle of mass 2 kg moves in simple harmonic motion and its potential energy U varies with position x as shown. The period of oscillation of the particle in seconds is (take $\pi = 3.14$)



A) 2.51 B) 1.15 C) 2.25 D) 0.25

18. A small magnet of moment 8 Am^2 is oriented along x-axis at the origin. A point P is at a distance of 2m from the centre of magnet and angle between OP and x-axis is $\tan^{-1}\left(\sqrt{2}\right)$. If $\mu_o = 4\pi\, x10^{-7}$ SI units the magnetic induction at P in tesla is n times 10^{-8}. Then n is (take $\sqrt{2} = 1.414$)

A) 12.10 B) 15.1 C)14.14 D) 12.12

19. The binding energies per nucleon for deuteron $\left({}_1H^2\right)$ and helium $\left({}_2He^4\right)$ are 1.1 MeV and 7.0 MeV, respectively. Calculate the energy released when two deuterons fuse to form a helium nucleus $\left({}_2He^4\right)$ in MeV.

A) 21.10 B) 23.60 C) 22.00 D)24.11

20. Electric and magnetic field of an electromagnetic wave are

A) in phase and perpendicular to each other

B) in phase and parallel to each other

C) in opposite phase and parallel to each other

D) in opposite phase and perpendicular to each other

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

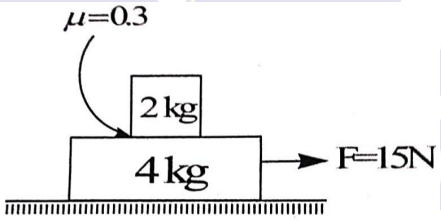
21. The displacement y of a particle executing simple harmonic motion is given by $y = 4\cos^2\left(\frac{t}{2}\right)\sin(1000t)$. This expression may be considered to be a result of the superposition of how many simple harmonic motions?

22. Three capacitors of capacities $8\mu F, 12\mu F, 16\mu F$, are connected in series. If the potential on $12\mu F$ is 60 V, the potential difference between the ends of the combinations is

23. A planet revolves about the sun in an elliptical orbit. The areal velocity (dA/dt) of the planet is 4.0×10^{16} m^2 / s. The least distance between planet and the sun is 2×10^{12} m. Then the maximum speed of the planet in km/s is

24. When a d.c. voltage of 200V is applied to a coil of self inductance $\frac{2\sqrt{3}}{\pi} H$, a current of 1A flows through it. But by replacing d.c. source with a.c source of 200V, the current in the coil is reduced to 0.5A. The frequency of a.c. supply in Hz is

25. Find the frictional force between the blocks in the figure. No friction between 4kg block and ground. $(g = 10\, ms^{-2})$



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SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Among the following the diamagnetic complexes is (are)

(i) $Ni(CO)_4$ (ii) $[Ni(PPh_3)_2Cl_2]$

(iii) $(NH_4)_2[PtCl_4]$ (iv) $[Pd(PPh_3)_2Cl_2]$

A) i, iii only B) ii, iii only C) i, iii, iv only D) all

27. Correct IUPAC name of $[Ag(NH_3)_2][Ag(CN)_2]$ is

(A) Diammineargentate (I) dicyanidoargentate (I)

(B) Diammineargentate (I) dicyanido silver (I)

(C) Diammine silver (I) dicyanidoargentate (I)

(D) Diammine silver (I) dicyanito silver (I)

28. The number of S-S bonds and $d_\pi - P_\pi$ bonds in cyclic trimeric SO_3, respectively, are :

A) 3, 6 B) 6,6 C) 0, 3 D) 0, 6

29. Brown ring formed in the test of nitrate ion is of:

A) $[Fe(H_2O)_6]SO_4$ B) $[Fe(NO)(CN)_5]SO_4$

C) $[Fe(NO)(H_2O)_5]SO_4$ D) $[Fe(NO)_5(H_2O)]SO_4$

30. In qualitaive analysis when H_2S is passed through an aqueous solution of salt acidified with dil. HCl, a black ppt is obtained. On boiling the ppt with dil. HNO_3, it forms a solution of blue color. Addition of excess of aqueous solution ammonia to this solution gives:

A) deep blue ppt of $Cu(OH)_2$

B) deep blue solution of $[Cu(NH_3)_4]^{2+}$

C) deep blue solution of $Cu(NO_3)_2$

D) deep ppt solution of $Cu(NO_3)_2.Cu(OH)_2$

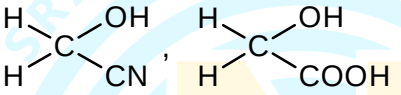
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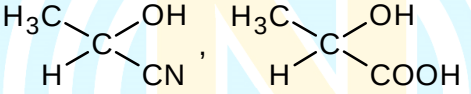
31. In which of the following compounds presence of nitrogen cannot be identified by Lassaigne's test ?

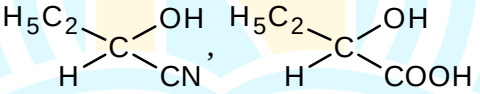
(i) hydrazine (ii) 2,4- DNP (iii) Urea (iv) Azobenzene

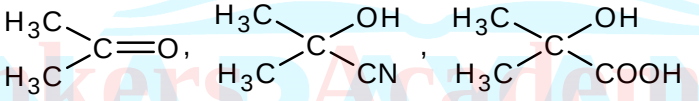
A) ii, iv B) i, iii, iv C) i, iv D) i, iii

32. A carbonyl compound 'A' reacts with hydrogen cyanide to form a Cyanohydrin 'B' which on hydrolysis gives an optically active alpha hydroxy acid 'C'. If 'A' gives a positive halo form test, then A,B and C are respectively.

A) HCHO, 

B) CH_3CHO, 

C) CH_3CH_2CHO, 

D) 

33. Identify the **correct** order(s) regarding the Group 13 elements.

I. First ionization enthalpy (B>Al).

II. First ionization enthalpy (Al>Ga).

III. First ionization enthalpy (Ga>In).

IV. First ionization enthalpy (In>Tl).

A) I, III B) II, IV C) I, II, III D) I, II, III, IV

34. Number of P-H bonds in hypophosphorous acid and number of P-OH bonds in ortho phosphorous acid respectively are

(A) 1,1 (B) 1,2 (C) 2,1 (D) 2,2

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35. Calculate the energy of photon involved during a transition from n=5 state to n=2 state in the hydrogen atom.

(A) $-4.58\times10^{-18}\,J$ (B) $-4.58\times10^{-19}\,J$

(C) $-4.58\times10^{-20}\,J$ (D) $-4.58\times10^{-17}\,J$

36. Which of the following oxide is amphoteric in nature

(A) N_2O (B) BaO (C) CO_2 (D) As_2O_3

37. The average composition of portland cement is

(A) $CaO(40-50\%), SiO_2(30-40\%),$

$Al_2O_3(5-10\%) MgO(2-3\%), Fe(1-2\%) SO_3(1-2\%)$

(B) $CaO(50-60\%), SiO_2(20-25\%), Al_2O_3(5-10\%), MgO(2-3\%),$

$Fe_2O_3(1-2\%), SO_3(1-2\%)$

(C) $CaO(40-50\%), SiO_2(20-25\%), Al_2O_3(2-3\%), MgO(5-10\%),$

Fe_2O_3 & $SO_3(1-2\%)$

(D) $CaO(50-60\%), SiO_2(30-40\%), Al_2O_3(2-3\%), Mg(5-10\%),$

Fe_2O_3 & $SO_3(1-2\%)$

38. Which of the following molecule(s) show(s) diamagnetic behavior?

(Consider s-p mixing wherever applicable)

I) B_2 II) C_2 III) N_2 IV) O_2 V) F_2 VI) S_2

A)IV,VI B)I,IV,VI C)II,IV,VI D)II,III,V

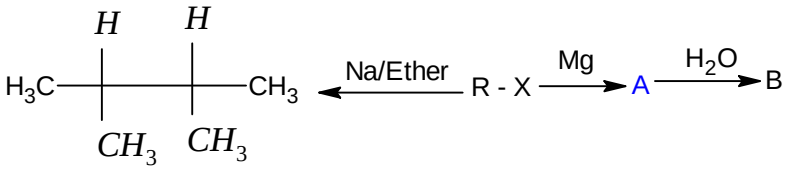
39. Calculate the enthalpy change on freezing of 1.0 mol of water taken at 10.0^0C to ice at -10.0^0C $\Delta H_{fus} = 6.03KJmol^{-1}$ at 0^0C. $C_P(H_2O)_{(l)} = 75.3J\,mol^{-1}K^{-1}$, $C_P = (H_2O)_{(S)} = 36.8J\,mol^{-1}K^{-1}$.

(A) $-71.51KJmol^{-1}$ (B) $-71.51Jmol^{-1}$

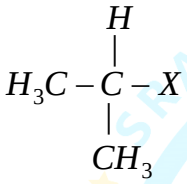
(C) $-7.151KJmol^{-1}$ (D) $-7.151.J.mol^{-1}$

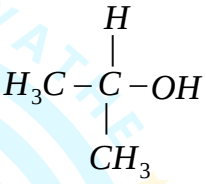
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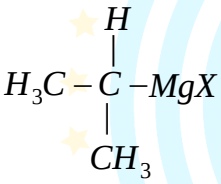
40. Identify the 'B' in the following reaction

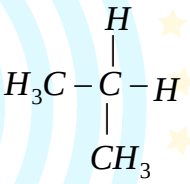


As one of the product

(A) 

(B) 

(C) 

(D) 

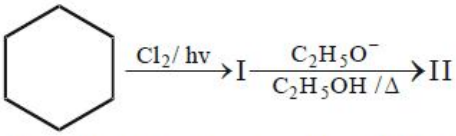
41. What are the hydrolysis products of Lactose?

(A) D-Galactose, D-Mannose (B) D-Galactose, D-Galactose

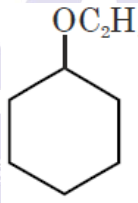
(C) D-Galactose, D-Glucose (D) D-Glucose, D-Glucose

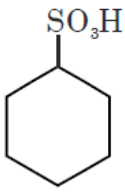
42.

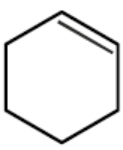
Consider the given sequence of reactions

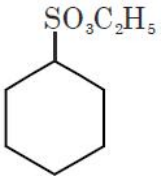


The final major product (II) is

A) 

B) 

C) 

D) 

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43. Arrange the following in increasing order of their basic strength

a) $C_2H_5NH_2$ b) $C_6H_5NH_2$

c) NH_3 d) $C_6H_5CH_2NH_2$

e) $(C_2H_5)_2NH$

(A) $b < c < d < e < a$ (B) $b < c < a < d < e$

(C) $b < c < d < a < e$ (D) $b < c < a < e < d$

44 Identify the 'E' in the following reaction

$$CH_3COOH \xrightarrow{NaOH} A \xrightarrow{Electrolysis} B \xrightarrow[(h\upsilon)]{Cl_2(1mole)} C \xrightarrow{alcKOH} D \xrightarrow{Br_2} E$$

(A) $C_2H_3Br_3$ (B) C_2H_4

(C) $\begin{matrix} CH_2Br \\ | \\ CH_2Br \end{matrix}$ (D) C_2H_5Br

45. The initial concentration of N_2O_5 in the following first order reaction, $N_2O_{5(g)} \rightarrow 2NO_{2(g)} + \frac{1}{2}O_{2(g)}$ was $1.24\times10^{-2} molL^{-1}$ at $318K$. The concentration of N_2O_5 after 60 minutes was $0.20\times10^{-2} molL^{-1}$. Calculate the rate constant of the reaction at $318K$ (given $\log 6.2 = 0.79$)

(A) $0.304\,min^{-1}$ (B) $0.03032\,min^{-1}$

(C) $0.0304\,sec^{-1}$ (D) $0.304\,sec^{-1}$

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. Number of milli moles of potassium permanganate required to react with 60 mL of 0.1 M hydrogen peroxide in faintly alkaline medium is $x\times10^{-2}$ then x is

47. What is the pH of the resultant solution when equal volumes of 0.1 M HCl and 0.2 M NH_3 aqueous solutions are mixed? Given pKb of NH_3 is 5.

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48. In Ostwald's process for the manufacture of nitric acid the first step involves the oxidation of NH_3 gas by oxygen gas to give nitric oxide gas and steam. What is the maximum weight of nitric oxide in grams that can be obtained, starting only with 10.00 gm of NH_3 and 20.00gm of oxygen?

49. The electrolysis of 250 mL of a NaCl solution was carried out for a period of 20 minutes with a current of 2.0 A. The resulting solution was titrated with 0.62 M HNO_3 solution. The volume of HNO_3 solution (in mL) required to achieve equivalence point is ___. (in nearest integer)

50. The hydrocarbon (X) with molar mass $54g\,mol^{-1}$ and 88.89% carbon has……..degree of unsaturation

MATHEMATICS **MAX.MARKS: 100**

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If a, b and c are three numbers (not necessarily different) chosen randomly and with replacement from the set $\{1,2,3,4,5\}$, then the probability that $(ab+c)$ is even, is

A) $\frac{35}{125}$ B) $\frac{59}{125}$ C) $\frac{64}{125}$ D) $\frac{75}{125}$

52. Let $y=f(x)$ and $f:R\to R$ be an odd function which is differentiable such that $f'''(x)>0$ and $\phi(\alpha,\beta)=\sin^8\alpha+\cos^8\beta+2-4\sin^2\alpha\cos^2\beta$. If $f''(\phi(\alpha,\beta))=0$ then $\sin^2\alpha+\sin^2\beta=$

A) 0 B) 1 C) 2 D) 3

53. If the domain of the function $f(x)=\frac{\sqrt{x^2-25}}{(4-x^2)}+\log_{10}(x^2+2x-15)$ is $(-\infty,\alpha)\cup[\beta,\infty)$, then $\alpha^2+\beta^2$ is equal to

A) 140 B) 150 C) 125 D) 175

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54. If the eccentricity of the ellipse $\frac{x^2}{a^2+1}+\frac{y^2}{a^2+2}=1$ *be* $\frac{1}{\sqrt{6}}$ then length of the latus rectum of the ellipse is

A) $\frac{5}{\sqrt{6}}$ B) $\frac{10}{\sqrt{6}}$ C) $\frac{8}{\sqrt{6}}$ D) $\frac{5}{\sqrt{2}}$

55. If the curve $3\sin y+12\sin^3 x=a$ does not intersect the line $y=3x$, then

A) $a\in(-\infty,-9)\cup(9,\infty)$ B) $a\in[-9,9]$

C) $a\in\{-9,9\}$ D) None of these

56. Let $\vec{a}=\cos\alpha\vec{i}+\sin\alpha\vec{j}$, $\vec{b}=\cos\beta\vec{i}-\sin\beta\vec{j}$, $\vec{c}=\cos r\vec{i}+\sin r\vec{j}$, $\vec{d}=\cos\delta\vec{i}-\sin\delta\vec{j}$. If $2\vec{a}+6\vec{b}+7\vec{c}+9\vec{d}=\vec{0}$, then

A) $3\cos(\alpha+\delta)=5\cos(\beta+r)$ B) $3\cos(\alpha+\delta)=7\cos(\beta+r)$

C) $\cos(\alpha+\delta)+\cos(\beta+r)=0$ D) $\sin(\alpha+\delta)+\sin(\beta+r)=0$

57. $f(x_1x_2)=f(x_1)f(x_2)\forall x_1,x_2\in R$ where $y=f(x)$ is a differentiable function, then which of the following must be true ?

A) $f(1)=0$ B) $f(0)=0$

C) $[f(x)]^2=[f(-x)]^2$ D) $f(x)=f(-x)$

58. The plane $2x+5y-4z-6=0$ and the line $\frac{x-1}{3}=\frac{y-2}{4}=\frac{z+2}{3}$, then equation of image of the given line with respect to the given plane is

A) $\frac{x+2}{79}=\frac{y+2}{40}=\frac{z+5}{-247}$ B) $\frac{x+2}{79}=\frac{y+2}{40}=\frac{z+5}{247}$

C) $\frac{x+1}{2}=\frac{y-2}{5}=\frac{z+2}{-4}$ D) $\frac{x+2}{2}=\frac{y+2}{5}=\frac{z+5}{4}$

59. let $z=\frac{-1+\sqrt{3}i}{2}$, where $i=\sqrt{-1}$ and $r,s\in\{1,2,3\}$. The number of ordered pairs (r, s) such that two vectors $(-z)^r\hat{i}+z^{3s}\hat{j}$ & $z^r\hat{i}+\hat{j}+z^r\hat{k}$ are perpendicular is __

A) 1 B)3 C)4 D) 6

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60. The total number of distinct $x \in [0,1]$ for which $\int_0^x \frac{e^t}{t+1}dt = \frac{(x+1)^2}{4}$ is____

A) 2 B) 1 C) 4 D) 3

61. $a_n = \int_0^{\frac{\pi}{2}} \frac{1-\cos 2nx}{1-\cos x}dx$, then $f(x) = \begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix} \forall x \in R;$, then $f(1) =$ __

A) 0 B) 2 C) 1 D) 3

62. If $\frac{4}{6x^2-10xy-9x+15y} = \frac{7}{5y^2-7yz-10y+14z} = \frac{7}{21z^2-9zx+7z-3x}$; $x, y, z \in R$, then

$\left|\frac{2x+1}{2x-3} + \frac{2y+3}{y-2} + \frac{3z+8}{3z+1}\right| =$_____

A) 0 B) 18 C) 4 D) 6

63. A function f is continuous for all x. (and not everywhere zero) such that $f^2(x) = \int_0^x f(t).\frac{\cos t}{2+\sin t}.dt$, then $f(x)$ is equal to

A) $\frac{1}{2}\ln\left(\frac{x+\cos x}{2}\right)$ B) $\frac{1}{2}\ln\left(\frac{3}{2+\cos x}\right)$

C) $\frac{1}{2}\ln\left(\frac{2+\sin x}{2}\right)$ D) $\frac{\cos x+\sin x}{2+\sin x}$

64. ${}^{10}C_0.{}^{20}C_{10} - {}^{10}C_1.{}^{18}C_{10} + {}^{10}C_2.{}^{16}C_{10} - {}^{10}C_3.{}^{14}C_{10} + \ldots\ldots\ldots =$

A) 0 B) 1 C) 2^{10} D) 3^{10}

65. The sum of the series $\cot^{-1}2 + \cot^{-1}8 + \cot^{-1}18 + \cot^{-1}32 + \ldots\ldots$ is

A) $\frac{\pi}{2}$ B) $\frac{\pi}{4}$ C) $\frac{\pi}{6}$ D) None of these

66. Let $S = \{1,2,3,\ldots.n\}$ and $A = \{(a,b): 1 \le a,b \le n\} = S \times S$. A subset B of A is said to be a good subset if $(x,x) \in B$ for every $x \in S$. Then the number of good subsets of A is

A) 1 B) 2^n C) 2^{n^2} D) $2^{n(n-1)}$

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67. If $x \in \left(0, \frac{\pi}{2}\right)$, one solution of $\frac{\sqrt{3}-1}{\sin x} + \frac{\sqrt{3}+1}{\cos x} = 4\sqrt{2}$ is $\frac{\pi}{12}$, the other solution must be $\frac{\lambda\pi}{36}$, then λ is

A) 11 B) 12 C) 13 D) 14

68. The value of $\int_{e^{-1}}^{e^2} \left|\frac{\log_e x}{x}\right| dx$ is

A) 3/2 B) 5/2 C) 3 D) 5

69. Let $\left(1+x^2\right)^2 (1+x)^n = \sum_{k=0}^{n+4} a_k x^k$. If a_1, a_2, a_3 are in AP, then n cannot be (given that ${}^nC_r = 0$, if $n < r$)

A) 6 B) 4 C) 3 D) 2

70. If $n > 0$ and $[.]$ denotes greatest integer function, then the value of $\lim_{x\to\infty} \frac{\log_e x^n - [x]}{[x]}$ is____

A) -1 B) 0 C) 1 D) n

SECTION – II
(Numerical Value Answer Type)

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71. If $\sum_{j=1}^{n-1} \frac{1}{1-e^{\frac{2\pi ij}{n}}} = \frac{n-1}{k}$, where $i = \sqrt{-1}$, then the value of k is____

72. A circle with centre (3, 0) has radius "r". As "r" varies no. of common tangents to the circle and the parabola $y^2 = 4x$ is denoted by f(r). Then $\sum_{r=1}^{10} f(r) =$ ___; $r \in N$

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73. Equation of chord of minimum length passing through (1, 2) of circle $x^2 + y^2 - 4x - 2y - 4 = 0$ is $ax + by + c = 0$ then $\left[\left|\frac{c}{b}\right|\right]$ is equal to ([.] denotes greatest integer function)

74. if $C_k = {}^6C_k$, then the value of $\sum_{k=1}^{6} k^3 \cdot \left(\frac{C_k}{C_{k-1}}\right)^2$ is

75. The number of distinct terms in the expansion of $\left(x + y^2\right)^{13} + \left(x^2 + y\right)^{14}$ is

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